SU(3) constants and conditional spectral comparisons

With the goal narrowed to $SU(3)$, each bound proved in these notes becomes a number. The group data are

$$N = 3, \quad \dim G = 8, \quad C_2(R_{\rm f}) = \frac{N^2-1}{2N} = \frac{4}{3}, \quad C_2(\text{adj}) = N = 3, \quad Z(G) = \mathbb{Z}_3,$$

in the normalization $\operatorname{tr} T^a T^b = \frac{1}{2}\delta^{ab}$, and the one- and two-loop coefficients are

$$b_0 = \frac{11N}{48\pi^2} = \frac{11}{16\pi^2} = 0.06966, \qquad \frac{b_1}{2b_0^2} = \frac{51}{121} = 0.4215.$$

The tables specialize group factors, running coefficients and stated conditional bounds. The Kogut–Susskind stability threshold remains existential. One-loop running to order-one coupling is an extrapolation. A $SU(3)$ field gap is not obtained by substituting group constants into the proved $SU(2)$ matrix theorem. Locality upper bounds and observed-channel estimates retain their separate scope.

1. Group data

quantity	value	where used
N	3	everywhere
$\dim G$	8	free-limit species count
$C_2(R_{\rm f})$	4/3	flux-loop energy, T2 gap
$C_2(\text{adj})$	3	Jacobian curvature factor
centre	$\mathbb{Z}_3$	flux sectors, valley minima
tr(1) in $R_{\rm f}$	3	plaquette norm $\|w_p\| \le 4N\hbar c/(ag^2) = 12\hbar c/(ag^2)$
b_0	$11/(16\pi^2) = 0.06966$	running, step count
$b_1/(2b_0^2)$	$51/121 = 0.4215$	the power in $a\Lambda_{\rm lat}$

The lattice Λ parameter of the obligations map §5 is therefore

$$a\Lambda_{\rm lat}(g) = (b_0 g^2)^{-51/121}\, e^{-1/(2b_0 g^2)}\,[1 + O(g^2)], \qquad \frac{1}{2b_0} = \frac{8\pi^2}{11} = 7.180.$$

2. The proved bounds, evaluated

T1. Unchanged in form: unique positive physical ground state and $\delta(g; N_s, SU(3)) > 0$ at every finite lattice and coupling.

T2 (T2 note). The perturbation parameter is

$$\beta = \frac{48N^2}{g^4(N^2-1)} = \frac{48\cdot 9}{8\, g^4} = \frac{54}{g^4},$$

so the criterion $\beta \le \beta_*(3, \{0,1\}^3)$ becomes

$$g \;\ge\; g_0 = \Big(\frac{54}{\beta_*}\Big)^{1/4},$$

and the gap is

$$\Delta_{a,L} \;\ge\; \gamma\, \frac{g^2}{2}\, \frac{4}{3}\, \frac{\hbar c}{a} = \frac{2\gamma}{3}\, g^2\, \frac{\hbar c}{a}.$$

A single plaquette flux-loop excitation in the fundamental, costs $2C_2 g^2 \hbar c/a = \frac{8}{3} g^2 \hbar c/a$.

Feynman–Bijl, strong coupling (upper-bound note Corollary 3). At $g = \infty$ the Haar variance is $\mathrm{Var}(\mathrm{Re}\,\mathrm{tr}\,U_p) = \frac{1}{2}$ for $N \geq 3$, so the two-sided statement reads

$$\frac{2\gamma}{3} g^2 \frac{\hbar c}{a} \;\leq\; \Delta_{a,L}^{\mathrm{phys}} \;\leq\; 12\, g^2 \frac{\hbar c}{a}$$

for g beyond both thresholds, a ratio of $18/\gamma$ between the two sides.

Lieb–Robinson (note). Four plaquettes per link and $\|w_p\| \leq 12\hbar c/(ag^2)$ give $J \leq 48\hbar c/(ag^2)$ and $v_{\mathrm{LR}} \leq 48 C_{\mathrm{LR}} c/g^2$. The former $96e\, c/g^2 \simeq 261 c/g^2$ uses the unverified $C_{\mathrm{LR}} = 2e$ normalization. It implies no actual superluminal speed. The gap gives $\hbar v_{\mathrm{LR}}/\Delta \leq 72 C_{\mathrm{LR}} a/(\gamma g^4)$; a clustering length also retains microscopic range.

Exact identities (note). With $C_2 = 4/3$ and $N = 3$,

$$\Delta V = \frac{16}{3}(3|\mathcal{P}| - V), \qquad |\nabla V|^2 \leq 16NV = 48\, V,$$

and the ground-state sum rule of that note holds with $2C_2 = 8/3$.

Blocking (criterion note). The block criterion is $\beta_{\mathrm{block}} = 72M^2/(g^2\delta(g;M))$ and the direct one is $\beta_{\mathrm{direct}} = 72/g^4$, from $32N/C_2 = 72$, so the ratio at strong coupling is $3M^2/8$ as before, blocking losing by $3/2$ already at $M = 2$.

Small volume. A spatially constant $SU(3)$ sector has three vectors in $\mathbb{R}^8$ with the $SU(3)$ structure constants. Homogeneity gives the candidate energy scale $g^{2/3}\hbar c/L$ if that matrix Hamiltonian has a positive first gap. The explicit matrix proof is for $SU(2)$. The $SU(3)$ spectral input and the full field theory's nonzero-mode comparison are additional obligations, not proved group substitutions.

3. The step count

With $2b_0 \log 2 = 2(0.06966)(0.6931) = 0.09657$, the number of renormalization doublings needed to run from a bare coupling to a strong-coupling threshold is

$$n = \frac{1}{0.09657}\Big(\frac{1}{g_{\mathrm{UV}}^2} - \frac{1}{g_{\mathrm{thr}}^2}\Big).$$

g_{UV}^2	n to $1/g_{\mathrm{thr}}^2 \to 0$	scale ratio 2^n
1	10.4	1.4×10^3
1/2	20.7	1.7×10^6
1/4	41.4	2.9×10^{12}

For $SU(2)$ the same table reads 15.5, 31.1, 62.1, since b_0 is smaller by $2/3$. **This one-loop extrapolation gives about a third fewer steps for $SU(3)$ than $SU(2)$** under the same bare-coupling convention. It is not a certified blocking trajectory or an assessment of total proof difficulty.

4. Two features specific to $SU(3)$

The defining representation is complex. For $SU(2)$, $\mathrm{tr}\,U_p$ is real, and the plaquette term is the full trace; for $SU(3)$ it is complex and the Wilson term keeps $\mathrm{Re}\,\mathrm{tr}\,U_p$. Consequences: the bound $\int (\mathrm{tr}\,U)^2 dU = 0$ used in the upper-bound note Corollary 3 holds for $N \geq 3$ and gives Haar variance $\frac{1}{2}$ rather than 1; and charge conjugation is a nontrivial symmetry, so the spectrum splits into C-even and C-odd sectors, and a C-even glueball insertion probes its own channel. This does not prove that the channel contains the global lowest excitation.

The centre is $\mathbb{Z}_3$. The valley minima of the valley note §4 lie at holonomies that are central, so there are three per direction and 27 electric-flux sectors on a three-torus, against 8 for $SU(2)$. These labels and central holonomy minima do not determine spectral weights or prove a physical gap. The quoted $O(\hbar c/L)$ barrier is the scale of the specified one-loop valley calculation.

5. Scope and consequence for STATE

The spectral obligations establish neither an exhaustive no-go classification of Hamiltonian methods nor an equivalence between all-coupling positivity and absence of every phase transition. The susceptibility/moment argument needs a nonzero admissible Hamiltonian insertion and infrared moments. Four-dimensional flow finiteness alone supplies neither.

In the specified free Gaussian example the channel estimate $4.26\,\hbar c/\sqrt{8t}$ is unchanged when eight colour species multiply numerator and denominator. It is not an interacting physical gap bound without the observable, domain and limit premises.

The group constants survive. Kogut–Susskind and Wilson thresholds, formal running counts, matrix truncations, observed channels and physical continuum limits retain their individual hypotheses.